ACASA

AUTOMATED COMPLIANCE AUDITING INTEGRATED WITH AI

LUMS, Class of 2025

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MOTIVATION

Manual audits are slow, error-prone, and hard to scale. ACASA automates critical ISO 27001:2022 controls, ensuring consistent, automated compliance and visibility into your security posture, with zero guesswork.

TECHNOLOGIES

Stack: Next.js + Mongoose
OS Compatibility: Windows, macOS, Linux
Automation Tools: Nmap, Selenium, Python Scripts

TARGET AUDIENCE

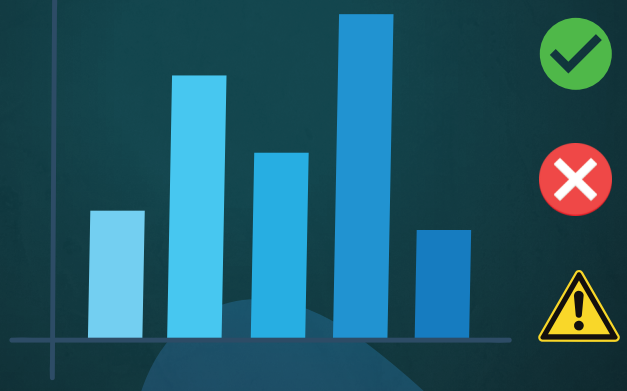
IT Admins & CISOs looking to simplify audit preparation
Security Consultants who manage multiple clients' compliance

BACKGROUND

Organizations today face ever-evolving cyber threats and growing regulatory obligations. ISO 27001's Annex A defines critical controls to secure systems, users, and data. ACASA bridges the gap between policy and implementation through intelligent automation. It ensures your company is compliant and audit-ready before approaching a certification authority — increasing trust, reducing risk, and giving clients confidence in how their data is handled.

FEATURES

Visualize Compliance Status



Automation of controls

Report Generation and notifications

Policy Generation

INSIGHT INTO AUTOMATED CONTROLS

ACASA automates compliance by converting ISO 27001 controls into actionable checks using Python scripts, system-level process monitoring, and Selenium-driven browser automation. It identifies endpoint risks, enforces access policies, and validates web filtering and employment compliance — all with minimal manual intervention and real-time feedback.

POLICIES

ISO policies translate company-specific information into tailored security controls that protect data and ensure compliance. Here is what we implemented:

- Privileged Access Policy
- Malware Detection Policy

FUTURE PLANS

We envision ACASA as a powerful, community-driven solution. By releasing it as open source on GitHub, we aim to make ISO 27001 compliance achievable for organizations of all sizes. Future plans include expanding coverage to all Annex A controls, integrating AI-based anomaly detection, and enabling seamless cloud deployment on AWS and Azure, making security smarter, faster, and more accessible.